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Generalized Regression Neural Networks Based HVDC Transmission Line Fault Localization

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***Abstract*—In this paper, a line fault location algorithm based on singular value decomposition and generalized regression neural networks (GRNN) is proposed for high voltage direct current (HVDC) transmission. As we know, the arriving instants of the first fault-induced transient backward travelling wave and the reflected wave can be detected. The high dimensional feature of travelling wave is different conditional on the term of diverse line fault position. Therefore, the fault distance can be estimated by using the velocity or current of the travelling wave. Firstly, we use singular value decomposition (SVD) method to extract the feature of HDVC travelling wave. After that, the features are sent to GRNN to model the relationship between the travelling wave and line fault position. For the sake of simplicity, the proposed algorithm is shorted for SVD-GRNN in the rest of this paper. Finally, simulation result indicates that line fault position can be accurately localized by the proposed algorithm.**

Keywords—HVDC; transmission line fault location; singular value decompostion; generalized regression neural networks

I. INTRODUCTION

With the rapid development of HVDC technology, HVDC transmission systems are widely explored and used in modern power systems as alternative to AC transmission systems. HVDC transmission systems with extra-long overhead lines such as the 2500km long Porto Velho-São Paulo system [1] and 295km long Basslink underground cables system [2], can be solutions for long distance high voltage, large capacity power transmission, even if asynchronous power system interconnection. Due to the reason that transmission line is one of the most important components of the electric power system, its protection is necessary for ensuring the stable and reliable operation of HVDC systems. Accurate fault location is needed for facilitating first-time restoration of the faulty line, and the detection of a transmission line fault enables swift isolation of the faulty line from service in avoid of the further harmful effects of the fault.

The use of travelling wave theory for fault detection was firstly proposed by Dommel and Michels in 1978. Since this method is suggested, considerable quantity of research effort has been devoted to developing travelling-wave-based methods for line fault localization in cable systems [2-6]. Traveling wave, which is a kind of non-stationary signal with mutation, spreads along the line when HVDC transmission line fault happens. The fault induced travelling waves comprise suffi-cient information with respect to the fault position, fault type, and sustain time. In AC power systems, the amplitude of a travelling wave stimulated by fault case mainly depends on the fault inception angle. Generally, the most widely used method in travelling wave protection and fault location in AC systems is the cross correlation method. From the aforementioned scheme, the similarity between an incident travelling wave and a subsequent backward travelling wave measured as the template signal is adopted as the evidence of the line fault localization. The correlation result approaches maximum value if the two signals have the same shape. Recently, some novel methods have been proposed to improve the accuracy of the line fault localization, such as wavelets based method [7], and pattern recognition based methods [8]. However, the accu-racy of fault location is still limited to the performance of the adopted feature extraction algorithm and pattern recognition algorithm. Generally, the non-linear fitting performance of the applied pattern recognition algorithm should be moderate in avoid of the fact of lack-fitting or over-fitting.

In this paper, the objective is to propose an GRNN based HVDC transmission line fault localization algorithm. By using this methodology for line fault localization, which is named SVD-GRNN for short in the rest of this paper, the feature of the traveling wave can be extracted from the voltage and current signals firstly by singular value decomposition. After the feature extraction, the feature vector is used as the training input sample of GRNN to mapping to the line fault location. Due to many researchers have validated that both current and voltage samples can be used for line fault localization, the proposed fault localization scheme is designed to work with only voltage samples as the system inputs. Finally, simulation is carried out using MATLAB/SIMULINK. A large number of fault data samples were generated for the validation. By this Monte Carlo simulation, the line fault localization accuracy of the proposed algorithm is verified.

II. MATHEMATICAL MODEL

Transmission line fault results in fault traveling wave, which bounces back and forth between the fault location and the system side. The fault traveling wave appears harmonic wave form with a series of specific frequency in the frequency domain. In order to analyze the fault location, Wavelet packet decomposition is adopted in this paper to extract the features of fault traveling wave value. With the aforementioned features, RBF neural network is used to achieve the function of pattern recognition. Finally, the transmission line fault position is presented by the output of the RBF neural network. The related mathematical model is demonstrated in this section.

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A. Singular Value Decomposition

As one of the most important tools of numerical analysis, particularly modem numerical linear algebra is the singular value decomposition. As indicated in [9], the SVD method was established for real square matrices originally by Beltrami and Jordan in the 1870's, and then for general rectangular matrices by Eckart and Young, which is known as the Autonne-Eokart-Young theorem. Sufficient research in literature verified that the SVD can be used to compute a number of the basic geometric objects of linear algebra reliably. This is but one of many important and fundamental fields of application. Considering the aforementioned excellent features of the SVD method, we adopt it in our work to extract the features of traveling wave with respect to voltage and current signals. This set of features is used to reflect the characteristic of HVDC transmission line fault case, and then the features are used as the training input sample of generalized regression neural networks to mapping to the line fault location.

Assuming metric $x(n)$ is one-dimensional time series of voltage traveling wave with numerical length of N, where $n = 1,2,\dots,N$. As discussed in the previous paragraph, the SVD method is operated only in two-dimensional state space. For extend the original signal to a two-dimensional matrix, which can be adopted to the SVD algorithm, we resample $x(n)$ by delay method. The resampling interval τ, and the reconstructed matrix of traveling wave $\mathbf{A}$ can be expressed as below:

$$\mathbf{A} = \begin{bmatrix} x(1) & x(1+\tau) & \cdots & x(1+(M-1)\tau) \\ x(2) & x(2+\tau) & \cdots & x(1+(M-2)\tau) \\ \vdots & \vdots & \ddots & \vdots \\ x(L) & x(L+\tau) & \cdots & x(N) \end{bmatrix} \tag{1}$$

where $N = L + (M-1)\tau$, $\mathbf{A}$ is a matrix by the size of $L \times M$. The order of $\mathbf{A}$ is r, which satisfies $r \leq min(L, M)$.

According to singular value decomposition theorem [10], there exists two orthogonal matrix, which is $\mathbf{U}$ and $\mathbf{V}$:

$$\mathbf{U} = [u_0, u_1, \dots, u_{r-1}] \in R^{L\times M}, \mathbf{U}^{\mathrm{T}} \cdot \mathbf{U} = \mathbf{I} \tag{2}$$

$$\mathbf{V} = [v_0, v_1, \dots, v_{r-1}] \in R^{L\times M}, \mathbf{V}^{\mathrm{T}} \cdot \mathbf{V} = \mathbf{I} \tag{3}$$

Besides the aforementioned orthogonal matrix $\mathbf{U}$ and $\mathbf{V}$, assumes a diagonal matrix, which is $\mathbf{\Lambda}$ satisfies:

$$\mathbf{\Lambda} = diag[\lambda_0, \lambda_1, \dots, \lambda_{r-1}] \in R^{L\times M} \tag{4}$$

$$\lambda_0 \geq \lambda_1 \geq \cdots \geq \lambda_{r-1}$$

Then the three matrix $\mathbf{U}$, $\mathbf{V}$ and $\mathbf{\Lambda}$ satisfies the relationship as below:

$$\mathbf{A} = \mathbf{U} \cdot \mathbf{\Lambda}^{\frac{1}{2}} \cdot \mathbf{V}^{\mathrm{T}} \tag{5}$$

In Eq. (4), metric λ_i is the non-zero feature value of matrix $\mathbf{A} \cdot \mathbf{A}^{\mathrm{T}}$ and $\mathbf{A}^{\mathrm{T}} \cdot \mathbf{A}$, where $\forall i \in [0,1,\dots,r-1]$. In Eq.(2) and Eq. (3), metrics u_i and v_i are the characteristic vector corresponding to λ_i with respect to matrix $\mathbf{A} \cdot \mathbf{A}^{\mathrm{T}}$ and $\mathbf{A}^{\mathrm{T}} \cdot \mathbf{A}$. Here we obtain the singular value of matrix $\mathbf{A}$, which is expressed as $\sqrt{\lambda_i}$. The aforementioned decomposition method is used as the feature extracted algorithm in this paper.

B. Generalized Regression Neural Networks

The mathematical principle of generalized regression neural networks is nonlinear regression analysis. GRNN has the capability of approximating any function, either linear or non-linear mathematical relationships between input and output variables. The advantage of GRNN is that it operates with sparse data in a real-time environment, because the regression surface is instantly defined. In other words, the learning technique of GRNN is different from the other popular learning method. It does not use an iterative process to tune approach to acquire the training result, but its learning process can be almost instantaneously finished. The GRNN algorithm stores the training sample sets and processes them by a nonlinear function to determine the output probability density functions. Therefore, the decision surfaces of the GRNN method are guaranteed to approach the Bayes-optimal decision boundaries with the increasing of the number of training samples.

By the principle of GRNN training algorithm, it can be used to model the relationship of the transmission line fault location and voltage travelling wave. The regression analysis of that the dependent variable $\mathbf{Y}$ with respect to dependent variable $\mathbf{X}$, is to search for the metric with the maximum probability. Assume the joint probability density function of random variable x and random variable y is $f(x, y)$. Providing the observed value of x is $\mathbf{X}$, then the regression of y with respect to $\mathbf{X}$, which is named as the conditional mean, is:

$$\widehat{\mathbf{Y}} = E(y/\mathbf{X}) = \frac{\int_{-\infty}^{\infty} y f(\mathbf{X}, y) \mathrm{d}y}{\int_{-\infty}^{\infty} f(\mathbf{X}, y) \mathrm{d}y} \tag{6}$$

Metric $\widehat{\mathbf{Y}}$ is the predicted output of $\mathbf{Y}$ condition on the input value of $\mathbf{X}$. According to Parzen Nonparametric Estimation method, the density function $\hat{f}(\mathbf{X}, y)$ can be estimated by sample set $\{x_i, y_i\}_{i=1}^{n}$. The calculation process can be shown as follow:

$$\hat{f}(X, y) = \frac{1}{n(2\pi)^{\frac{p+1}{2}} \sigma^{p+1}} \times \sum_{i=1}^{n} \left\{ exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}} \cdot (\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right] exp\left[-\frac{(\mathbf{X}-\mathbf{Y}_i)^2}{2\sigma^2}\right] \right\} \tag{7}$$

where $\mathbf{X}_i$ and $\mathbf{Y}_i$ are the observed values of random variable x and y; Metric n is sample size; Metric p is the dimension of random variable x; Metric σ is the spread factor of Gauss function, which is also named as smoothing factor.

We use $\hat{f}(\mathbf{X}, y)$ instead of $f(\mathbf{X}, y)$ in Eq.(7), and switch the sequence of integral and addition operation, and obtain:

$$\begin{cases} \widehat{\mathbf{Y}}(\mathbf{X}) = \dfrac{\sum_{i=1}^{n} \left\{ \vartheta \cdot exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}} \cdot (\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right] \right\}}{\sum_{i=1}^{n} \left\{ \varTheta \cdot exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}} \cdot (\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right] \right\}} \\ \vartheta = \displaystyle\int_{-\infty}^{\infty} y\, exp\left[-\frac{(\mathbf{Y}-\mathbf{Y}_i)^2}{2\sigma^2}\right] \mathrm{d}y \\ \varTheta = \displaystyle\int_{-\infty}^{\infty} exp\left[-\frac{(\mathbf{Y}-\mathbf{Y}_i)^2}{2\sigma^2}\right] \mathrm{d}y \end{cases} \tag{8}$$

Considering the fact that $\int_{-\infty}^{\infty} ze^{-z^2}\mathrm{d}z = 0$, process the two aforementioned Integral operation, and then we can obtain the output of network $\widehat{\mathbf{Y}}(\mathbf{X})$ as follow:

$$\widehat{\mathbf{Y}}(\mathbf{X}) = \frac{\sum_{\mathrm{i}=1}^{n} \mathbf{Y}_i \exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}}\cdot(\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right]}{\sum_{\mathrm{i}=1}^{n} \exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}}\cdot(\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right]} \tag{9}$$

The predicted value $\widehat{\mathbf{Y}}(\mathbf{X})$ is obtained by the weighted mean of the observed sample value $\mathbf{Y}_i$. The weight factor of every observed value $\mathbf{Y}_i$ is the Euclidean distance squared index of the corresponding sample $\mathbf{X}_i$ and $\mathbf{X}$. Metric $\widehat{\mathbf{Y}}(\mathbf{X})$ nearly approaches to the mean of the dependent variable, where the smooth factor σ approaches infinity. On the opposite, Metric $\widehat{\mathbf{Y}}(\mathbf{X})$ nearly approaches to the training sample, where the smooth factor σ approaches zero. The predictive value calculated by this equation could be very close to the corresponding dependent variable in training sample set, if the input value needs to be estimated is included in the prediction of the training set. However, once the input value failed to be included into the training sample, the accuracy of prediction could be very poor. This phenomenon indicates poor generalization ability of the network. The dependent variable of all the training samples can be included, if σ value is moderate. As a result, the dependent variable of training sample, which is close to the predicted value, obtains much weight.

In this paper, the adopted GRNN algorithm is composed by four layers, which are input layer, pattern layer, summation layer and output layer, respectively. The structure of GRNN is shown in Fig. 1. Assuming input vector is $\mathbf{X} = [x_1, x_2, \ldots, x_n]^{\mathrm{T}}$ and output vector is $\mathbf{Y} = [y_1, y_2, \ldots, y_k]^{\mathrm{T}}$, the function of each layer is shown as below:

1) input layer:

The number of neurons equal to the dimension of input vector and they achieve the function of transmiting the input vector directly to the pattern layer.

2) pattern layer:

The number of neurons equals to the number of training samples, and each neuron corresponding to a different sample. The transmitting function of pattern layer is:

$$p_i = exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}}(\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right], \quad i = 1,2,\ldots,n \tag{10}$$

The Euclidean distance squared index of the output of neuron i and its corresponding sample $\mathbf{X}$ is $D_i^2 = (\mathbf{X} - \mathbf{X}_i)^{\mathrm{T}}(\mathbf{X} - \mathbf{X}_i)$, where $\mathbf{X}$ is the input vector and $\mathbf{X}_i$ is the ith training sample related to ith neuron.

3) summation layer:

Generally, two types of neurons can be used in this layer to achieve the summation operation. The first type of function is:

$$\mathrm{S}_D = \sum_{i=1}^{n} p_i = \sum_{i=1}^{n} \exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}}\cdot(\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right] \tag{11}$$

This type of neuron performs arithmetic summation on the output of summation layer. The connective weight between pattern layer and summation layer is 1.

The other type of function is:

$$\begin{aligned}\mathrm{S}_{Nj} &= \sum_{i=1}^{n} y_{ij}\cdot p_i \\ &= \sum_{i=1}^{n} \mathbf{Y}_i \cdot exp\left[-\frac{(\mathbf{X}-\mathbf{X}_i)^{\mathrm{T}}(\mathbf{X}-\mathbf{X}_i)}{2\sigma^2}\right]\end{aligned} \tag{12}$$

By this type of function, the connective weight between the ith neuron of pattern layer and the jth neuron of summation layer is the jth element of the ith output layer $\mathbf{Y}_i$.

4) output layer:

The number of neuron in output layer equals to the dimension of output vector. Each neuron achieve division function on the output of summation layer, and the output of jth neuron corresponding to the jth element of the predicted $\widehat{\mathbf{Y}}(\mathbf{X})$.

$$y_i = \frac{S_{Nj}}{S_D} \quad, j = 1,2,\ldots,k \tag{13}$$

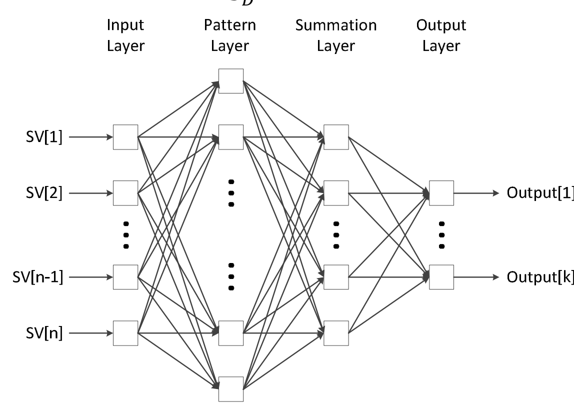

Fig. 1 the structure of RBF neural network

III. Simulation Result and Discussion

In this section, we investigate the performance of the proposed line fault location algorithm based on the HVDC transmission system model shown in Fig. 2. The system model is comprised by 12-pulse bridges, which is series connected by two 6-pulse bridges. The adopted HVDC transmission model has 500kV as the nominal dc voltage and 2kA as the nominal dc current. It is designed to deliver 1000mW of active power. Furthermore, a bipolar HVDC configuration is used since most of the present day HVDC systems are built in bipolar configuration, instead of the mono-polar arrangement in the original reference. A frequency dependent distributed parameter model of a 500kM long underground cable line is adopted as the transmission line. A series DC reactor is important in line commutated type HVDC system. Due to the fact that smoothing reactor is included in the simplified cable model, original test network does not contain a separate one. Considering the fact that typical value of the smoothing reactor is in the range of 0.5-1H, 0.5H smoothing reactor is placed in series with the transmission line at both ends as shown in Fig. 2.

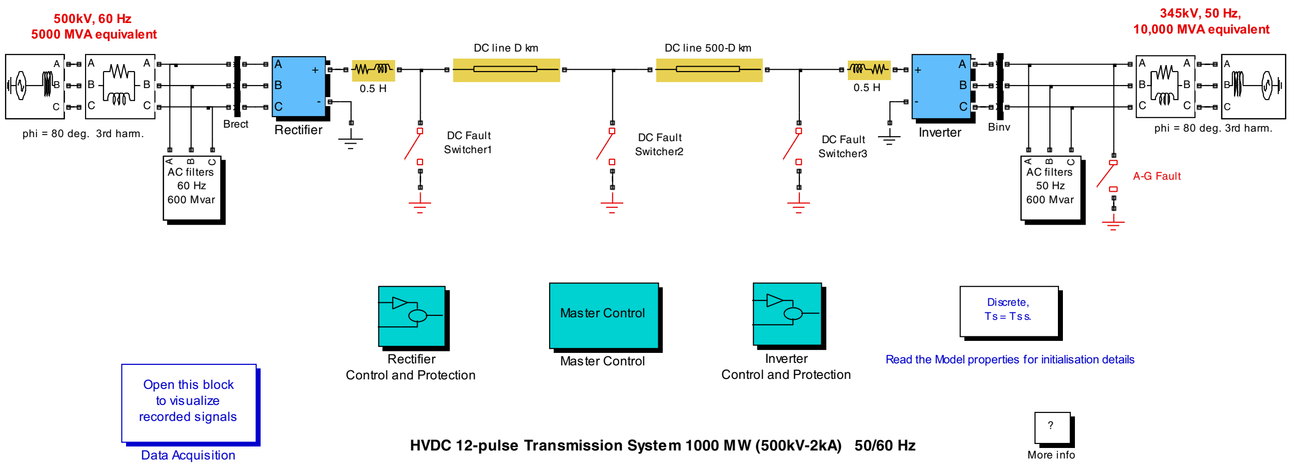

Fig. 2 HVDC simulation network modeled in MATLAB/SIMULINK

As shown in Fig. 2, we deploy three switchers alone the transmission line. For the sake of switcher 2, it achieves the function of controlling the line fault position. In the simulation, switcher 2 is set to be open initially, and it will be closed at timing of 500ms, which results in short circuits in the transmission line. We investigate the line fault cases that short circuits happen on the transmission line per 10kM. Base on this condition, 50 groups of line fault scenario are generated by SIMULINK. In Fig. 3, we select four typical traveling wave of voltage based on the condition of short circuits, which happen on 20km, 100km, 200km and 300km, respectively. From the figure we can observe the traveling wave appears different shape base on different line fault position. This phenomenon is the reason why we use SVD algorithm to extract the feature of the traveling wave.

The AC network capacity is 5000mVA by the rectifier side, and the system frequency is 60Hz. the inverter side AC grid connected 345kV system capacity is 10000mVA, and the system frequency is 50Hz. The capacity of the two AC filters in Fig. 2 is both 600mVAR. The AC filters not only include reactive power compensation equipment, which provides reactive power for converter, but also include a set of filter composed by capacitor set. AC filter directly parallel connected with the AC system line. The rectifier works in current control mode, and the inverter works in voltage control mode. The system is discretized in the simulation process, and the sampling frequency is 20kHz. For each group of line fault simulation, 100 sets of traveling wave are recorded and investigated. Therefore, 5000 sets of data in total are used in this simulation.

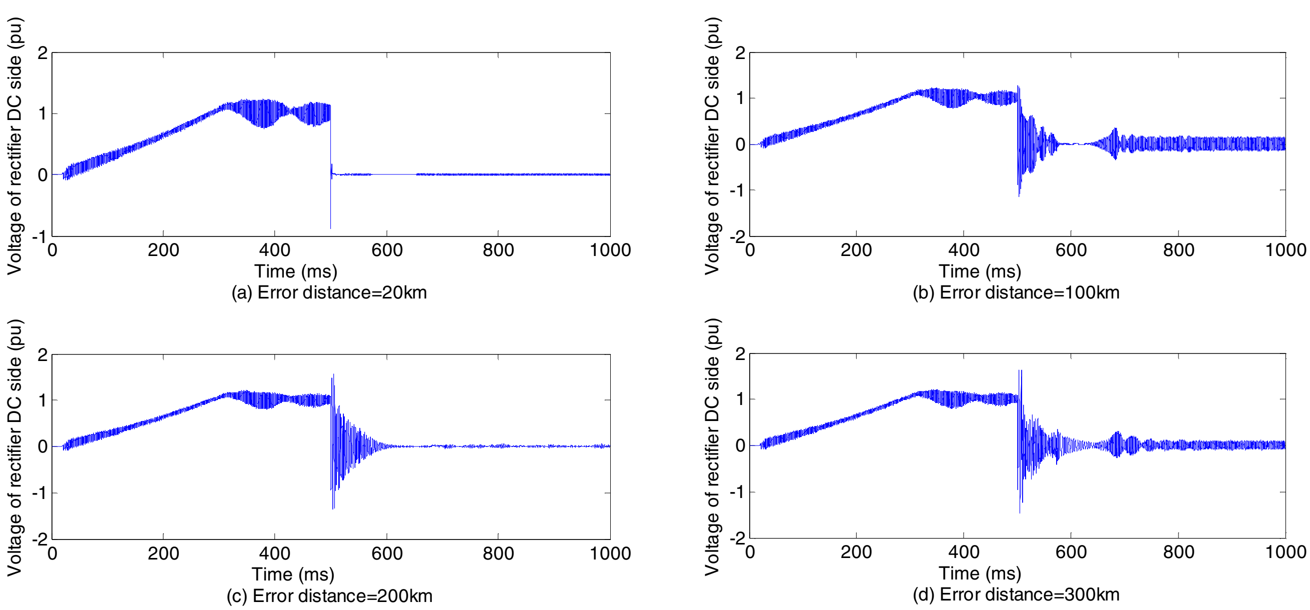

Fig. 3 Four typical traveling wave of voltage based on the condition of short circuits

The mean error of fault localization is presented to investigate the reliability of the research results. By the proposed SVD-GRNN line fault localization algorithm, simulation results are shown in. From Fig. 3 we can observe that the mean error of fault localization is less than 0.05kM approximately at each simulated line fault position. Fig. 5 indicates that, by the proposed SVD-GRNN algorithm, the mean cumulative distribution probability is about 65%, 85%, 95%, and 99% base on the condition that error of fault localization is less than 0.05kM, 0.1kM, 0.15kM, and 0.2kM, respectively. In conclusion, line fault position can be accurately localized by the proposed algorithm.

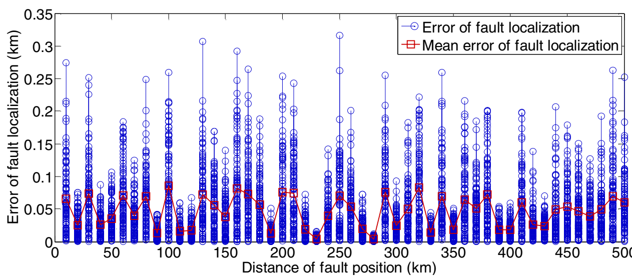

Fig. 4 Error of fault localization

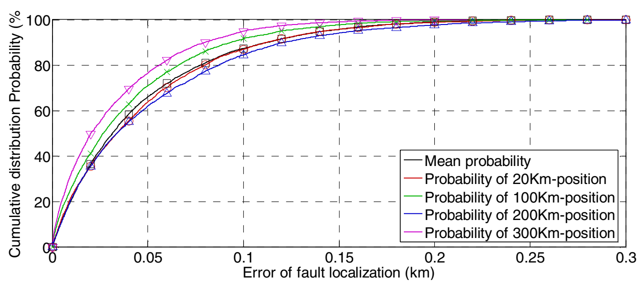

Fig. 5 Cumulative distribution probability of fault localization error

IV. Conclusions

We proposed an HVDC transmission line fault localization algorithm based on generalized regression neural networks with singular value decomposition in this paper. By adopting singular value decomposition method, the feature of the traveling wave can be extracted from the voltage and current signals, which can be used as the training input sample of generalized regression neural networks to model the relationship between traveling wave and line fault position. From the simulation result we can carry out the conclusion that, the mean error of fault localization is 0.05kM approximately at each tested line fault point. In conclusion, line fault position can be accurately localized by the proposed algorithm.

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